



Jyoti Prakash

Bachelor of Technology
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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in mobile application development, backend services, and AI applications. Built a React Native EdTech platform, developed backend features using Node.js, Express.js, PostgreSQL, and AWS S3, and implemented a Retrieval-Augmented Generation (RAG) assistant and a custom 64-bit RISC-V processor. Interested in scalable backend systems and AI-driven software development.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech, ECE	National Institute of Technology, Patna	8.11 CGPA	2027(Expected)
Senior Secondary	Teresa International Academy, Patna / CBSE	85.8%	2022
Secondary	Shivam School, Bihta / CBSE	92.8%	2020

EXPERIENCE

- **Astomverse** [\[Website\]](#) [\[Certificate\]](#) July 2024 – January 2025
VR Developer Intern Patna, India
 - Developed Unity3D and C# VR learning modules covering 4 interaction patterns: spatial navigation, quizzes, object interaction, and dynamic UI.
 - Reduced polygon count by 30% and improved frame rates by 25% across standalone and low-end VR headsets through asset and scene optimization.
 - Engineered 4 reusable C# component groups for teleportation, interaction, quiz flows, and UI overlays, enabling faster learning-module updates without core code changes.
 - Standardized cross-device input and rendering compatibility layers to improve reliability across multiple VR hardware targets.

PROJECTS

- **Prayas IAB – Scalable Content Platform (Freelance)**
Technologies: React Native, Expo, Redux Toolkit, Node.js, Express.js, PostgreSQL, Prisma, AWS S3
 - Developed a React Native EdTech platform with a six-level academic hierarchy supporting categories, classes, modules, subjects, chapters, quizzes, and video lessons.
 - Designed normalized PostgreSQL schemas using Prisma ORM and developed scalable RESTful APIs with Node.js and Express.js for authentication, content delivery, analytics, and progress tracking.
 - Implemented Redux Toolkit state management and role-based access control (RBAC) for three role-specific experiences: students, admins, and content managers.
 - Delivered a timed quiz engine with auto-scoring and AWS S3 cloud storage for protected solution-video streaming.
- **AI Knowledge Assistant – RAG-Powered Q&A System** [\[🔗\]](#)
Technologies: Python, FastAPI, React.js, Gemini API, Ollama, Qdrant, Hugging Face, Crawl4AI
 - Built a Retrieval-Augmented Generation (RAG) assistant using FastAPI, Gemini API, Ollama, Hugging Face embeddings, and Qdrant for document and web-based question answering.
 - Implemented a 6-stage retrieval pipeline: ingestion, chunking, embedding generation, Qdrant indexing, semantic search, and response generation.
 - Integrated Crawl4AI to extract, process, and index external web content into the vector database for expanded knowledge coverage.
 - Developed and documented REST APIs for document upload, website crawling, retrieval, and conversational chat, integrated with a React frontend.
- **64-bit RISC-V Processor with ML Accelerator** [\[🔗\]](#)
Custom RV64 processor with memory-mapped ML acceleration for inference and training.
 - **Technologies:** RISC-V, Verilog RTL, Icarus Verilog, GTKWave, Python
 - Implemented RV64I/M/B processor with decode, ALU, register file, control, and memory interface
 - Designed memory-mapped ML accelerator with SIMD int8 (8 MACs/cycle), dot-product/GEMM engines, and DMA; no ISA changes
 - Verified with 9,900+ differential checks against independently generated RISC-V reference models
 - Ran regression/MLP inference and fixed-point gradient training in integer arithmetic, reducing 784-feature workload from 3,100 to 48 instructions

CODING PROFILES

◦ **DSA Practice:** 350+ problems solved across [LeetCode](#) (250+), [GeeksforGeeks](#) (100+), and [CodeChef](#) (40+).

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript , C++, C, Verilog
- **Frontend & Mobile:** React.js, React Native, Expo, Redux Toolkit, HTML, CSS
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, JWT Authentication, Prisma
- **Databases:** PostgreSQL, MongoDB,
- **AI / LLM:** Retrieval-Augmented Generation (RAG), LLM Integration, Gemini API, Ollama, Hugging Face Embeddings, Semantic Search, Prompt Engineering
- **Cloud & Tools:** Git, GitHub, API Testing (e.g. Postman), Unity3D, Blender, GTKWave, Icarus Verilog

AWARDS & ACHIEVEMENTS

- **MSME Idea Hackathon 4.0 (2024):** Selected by the Ministry of MSME, Government of India, as a team member for an innovation project under the national idea hackathon.

CERTIFICATIONS

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| ◦ NPTEL (IIT Madras), The Joy of Computing using Python – Elite, 85% | 2025 |
| ◦ NPTEL (IIT Kharagpur), Cloud Computing – Elite, 82% | 2026 |
| ◦ NPTEL (IIT Guwahati), Digital Design with Verilog – Elite, 73% | 2026 |

LEADERSHIP & EXTRACURRICULARS

- **Team Lead, Astomverse Community** – Led a team of 10+ in building interactive VR simulations at NIT Patna.
- **Member, DFC Club** – NIT Patna 2024 – Present